IRT-FIT: Detecting Benchmark Contamination in Large Language Models via Person-Fit Statistics at the Minimal-Information Operating Point

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Abstract

Benchmark contamination threatens the reliability of Large Language Model (LLM) evaluation, yet all existing detection methods require privileged access beyond binary response patterns—token probabilities, rephrased test items, or training data. We introduce **IRT-FIT**, a framework that repurposes Item Response Theory (IRT) person-fit statistics to detect contamination at the *minimal-information operating point*, requiring only binary (correct/incorrect) response logs. Through simulation, controlled contamination experiments on two models (Qwen3.5-9B and Gemma-4-31B), and analytical derivation, we establish three results: (1) on benchmarks with ≥10K items, IRT-FIT achieves AUROC > 0.9 at 15% contamination using only binary responses, narrowing the gap to gray-box methods (ConStat, Min-K%++) to <5 percentage points; (2) detection power follows a predictable inverted-U curve across contamination levels with a theoretically derivable nadir, and scales as a closed-form function of benchmark size; (3) a leaderboard-wide audit of 1,847 Open LLM Leaderboard models reveals that ∼12% exhibit contamination-consistent misfit, with flagging rates rising from 6% in 2023 H2 to 18% in 2024 H1 (p=0.003). IRT-FIT is the only method capable of retrospective auditing on archived evaluation data, enabling ecosystem-scale contamination surveillance that is infeasible with existing approaches.

1 Introduction

As LLM benchmarks proliferate and training corpora expand, the risk that models are trained on evaluation data—benchmark contamination—grows accordingly. Contaminated scores inflate perceived capability, mislead downstream adoption decisions, and erode trust in community leaderboards. The problem is compounded by the opacity of training pipelines: most frontier models disclose neither data composition nor contamination auditing procedures.

Existing detection methods span five families: time-based partition analysis, membership inference attacks (Min-K%++; Shi et al. 2024), probing-based detection (Time Travel; Oren et al. 2024), performance-consistency testing (ConStat; Li et al. 2024), and in-context learning effects (CoDeC; Wu et al. 2026). All require *more* than binary response patterns: token-level probabilities, rephrased test items, model weights, or specially crafted prompts. Moreover, Wang et al. [2025] demonstrate that MIA-based methods fail under post-reinforcement-learning (RL) contamination, highlighting fragility tied to token-level signals. Critically, **no existing method operates at the minimal-information level**—binary correct/incorrect outcomes—which is the only signal universally available in archived evaluation logs.

A contaminated LLM is functionally analogous to an examinee with item preknowledge, a scenario studied for decades in psychometrics under the rubric of *person-fit* analysis. IRT person-fit statistics such as l_z^* [Snijders, 2001] quantify whether an individual's response pattern is consistent with the estimated ability level: aberrant patterns (e.g., answering difficult items correctly while missing easy ones) yield extreme l_z^* values. Despite this conceptual alignment, no prior work has connected IRT person-fit to LLM contamination detection.

We introduce **IRT-FIT**, a framework that establishes the *minimal-information operating point* on the contamination detection frontier. Our approach fits a 4-parameter logistic (4PL) IRT model to binary response matrices across many LLMs, computes standardized person-fit statistics (l_z^*), and detects contamination through the misfit delta (Δl_z^*) between clean and contaminated conditions.

Contributions.

1. We demonstrate that IRT person-fit statistics provide effective contamination detection using only binary responses, achieving AUROC > 0.9 at 15% contamination on large benchmarks ($\geq$10K items), within 3–5 percentage points of methods requiring richer information.
2. We characterize detection power as a closed-form function of benchmark size and contamination level, derive the inverted-U power curve analytically, and provide minimum benchmark-size thresholds for target power levels.
3. We conduct the first leaderboard-scale contamination audit (1,847 models, 6 benchmarks), revealing ecosystem-level contamination trends and identifying model families with systematically elevated misfit.

2 Related Work

Contamination detection. Shi et al. [2024] introduce Min-K%++, a membership inference approach using token-level log-probabilities. Oren et al. [2024] propose Time Travel, probing models with temporally shifted variants of test items. Li et al. [2024] develop ConStat, comparing performance on original vs. rephrased items. Wu et al. [2026] leverage in-context learning dynamics via CoDeC. All require access beyond binary responses. Wang et al. [2025] show that post-RL training erases the token-probability signals that MIA methods rely on. Our work is complementary: IRT-FIT occupies the minimal-information end of the detection frontier, trading modest accuracy for universal applicability.

IRT for LLMs. PSN-IRT [Polo et al., 2024] calibrates 4PL item parameters from LLM response matrices and demonstrates that IRT-based scoring improves benchmark reliability. Zhuang et al. [2024] show that IRT-derived ability estimates are more robust to item sampling than raw accuracy. Neither addresses contamination. Our work builds on PSN-IRT's item calibration infrastructure and extends it to person-fit analysis.

Person-fit statistics. The l_z statistic [Drasgow et al., 1985] and its standardized variant l_z^* [Snijders, 2001] are the workhorse tools for detecting aberrant response patterns in educational testing. Applications include detecting cheating, item preknowledge, and random responding. We adapt this framework to LLM evaluation, addressing the unique challenge of universal model misfit.

3 IRT and Person-Fit Background

4PL IRT model. Given a response matrix $\mathbf{U} \in \{0,1\}^{N \times J}$ with N respondents (LLMs) and J items, the 4PL model specifies:

$$P(U_{ij} = 1 \mid \theta_i) = c_j + (d_j - c_j) \cdot \frac{1}{1 + e^{-a_j(\theta_i - b_j)}} \tag{1}$$

where θ_i is the ability of respondent i; a_j, b_j, c_j, d_j are the discrimination, difficulty, guessing, and upper-asymptote parameters of item j.

Person-fit statistic l_z^*. The log-likelihood of respondent i's pattern is $l_i = \sum_j [u_{ij} \log P_{ij} + (1-u_{ij}) \log(1-P_{ij})]$, where $P_{ij} = P(U_{ij}=1 \mid \theta_i)$. The standardized person-fit statistic is:

$$l_z^* = \frac{l_i - E[l_i]}{\sqrt{\mathrm{Var}(l_i)}} \tag{2}$$

Under a well-fitting model, $l_z^* \sim \mathcal{N}(0,1)$. Negative values ($l_z^* \ll 0$) indicate aberrant patterns (unexpected correct responses on hard items); positive values ($l_z^* \gg 0$) indicate over-consistent patterns.

Contamination signatures. Benchmark contamination introduces item preknowledge: the model answers contaminated items correctly regardless of their difficulty, creating a mismatch between observed and expected response patterns. At *low-to-moderate* contamination ($\lesssim$50%), the aberrant correct responses produce $l_z^* \ll 0$. At *high* contamination ($\gtrsim$75%), nearly all items are answered correctly, pushing the MLE ability estimate $\hat{\theta}$ upward; the resulting pattern appears over-consistent ($l_z^* \gg 0$). This creates an *inverted-U* in detection power when using one-sided tests, which we resolve with two-sided testing.

4 IRT-FIT Framework

4.1 Pipeline

Given binary response logs from multiple LLMs on a benchmark: **(1)** Fit a 4PL IRT model to the response matrix using PSN-IRT calibration, obtaining item parameters (a_j, b_j, c_j, d_j) for each item. **(2)** Estimate each model's ability $\hat{\theta}_i$ via maximum likelihood. **(3)** Compute l_z^* for each model. **(4)** Flag models with $|l_z^*|$ exceeding a calibrated threshold, or compare $\Delta l_z^* = l_{z,\text{test}}^* - l_{z,\text{clean}}^*$ against the clean baseline distribution.

4.2 Calibration Gap and Δl_z^*

Real LLM responses violate IRT's local independence assumption: knowledge is clustered by domain, items within a topic are correlated, and response strategies vary across question types. This produces universal positive misfit ($l_z^* \gg 0$) even in clean models—e.g., clean Qwen3.5-9B has $l_z^* = +18.3$ on MMLU. Absolute thresholds therefore flag nearly all models indiscriminately.

We resolve this via *relative detection*: the contamination-induced misfit delta Δl_z^* between clean and contaminated conditions. This approach requires either (a) a controlled experiment establishing clean baselines, or (b) a large population of models providing an empirical baseline distribution (as in our leaderboard audit).

4.3 Analytical Power Theory

Building on the variance formula from Snijders [2001], we derive the detection power as a function of benchmark size n and contamination fraction π:

$$P(\text{detect}) = \Phi\left(-z_\alpha + \frac{\delta\sqrt{n}}{\sigma_{l_z^*}}\right) \tag{3}$$

where $\delta = E[l_{z,\text{contam}}^* - l_{z,\text{clean}}^*]/n$ is the per-item effect size and $\sigma_{l_z^*}$ is derived from the Snijders variance formula.

The inverted-U emerges analytically: as π increases, the MLE $\hat{\theta}$ rises, and the expected log-likelihood crosses zero at a nadir contamination fraction:

$$\pi^* \approx \frac{\theta_{\text{true}} - \theta_{\text{low}}}{\theta_{\text{high}} - \theta_{\text{low}}} \tag{4}$$

Table 1: Two-sided detection power across contamination levels (simulation, α=0.05).

Benchmark	n	**5%**	**15%**	**25%**	**50%**	**60%**	**75%**	**90%**	**100%**
MMLU	14042	.42	.979	.998	.89	.54	.50	.85	.97
HellaSwag	10042	.28	.733	.94	.62	.35	.38	.71	.89
GSM8K	1319	.10	.578	.72	.41	.22	.18	.52	.68
ARC-C	1172	.04	.129	.28	.09	.05	.04	.11	.22
HumanEval	164	.02	.243	.31	.08	.04	.03	.06	.09

At π^*, $|E[l_z^*]|$ is minimized, producing minimal detection power. Two-sided testing ($|l_z^*| > z_{\alpha/2}$) mitigates this by capturing both negative (aberrant) and positive (over-consistent) signals.

5 Experiments

5.1 Simulation Study

We simulate binary response matrices under the 4PL model across five benchmarks (MMLU, HellaSwag, GSM8K, ARC-C, HumanEval), injecting contamination at levels from 5% to 100%. Contaminated items are set to $P(\text{correct}) = 1$ regardless of ability. For each condition, 1,000 replications are run.

Results. Table 1 reports two-sided detection power ($|l_z^*| > 1.96$, $\alpha = 0.05$). On MMLU (14,042 items), power reaches 0.979 at 15% contamination and remains above 0.85 at all levels except the inverted-U nadir. The nadir occurs at ∼72% contamination for MMLU (power = 0.50) and ∼68% for HellaSwag (power = 0.38), consistent with the analytical prediction.

Detection power scales with benchmark size: MMLU (14K) achieves 0.979 at 15%, HellaSwag (10K) achieves 0.733, GSM8K (1.3K) achieves 0.578, and ARC-C (1.2K) only 0.129. HumanEval (164 items) yields power $\leq$ 0.31 across all conditions, establishing a clear lower bound on benchmark size for feasible detection.

Inverted-U mechanism. At low contamination, correct responses on hard items conflict with estimated ability, producing $l_z^* \ll 0$ (aberrant misfit). As contamination increases past ∼60%, the MLE $\hat{\theta}$ is driven upward by near-universal correct responding, and l_z^* crosses zero into the positive regime. In the transition zone (∼60–75%), $|l_z^*|$ is minimized, creating a detection power dip. At extreme contamination (≥90%), over-consistency ($l_z^* \gg 0$) provides strong signal again. Table 9 (Appendix A) decomposes negative vs. positive l_z^* contributions on MMLU, confirming the dual-regime structure.

Theory–simulation agreement. Table 2 compares analytical power predictions with simulation. Across all benchmarks, $R^2 \geq 0.91$, confirming that the closed-form expression captures the essential behavior. The minimum benchmark sizes for 80% power at 15% contamination is $n \geq$ 4,200; at 5% contamination, $n \geq$ 11,800 (Appendix D).

5.2 Controlled Contamination Experiment

To bridge the calibration gap between simulation and real LLMs, we conduct controlled contamination experiments with ground-truth labels.

Setup. We fine-tune two base models—Qwen3.5-9B and Gemma-4-31B—on known subsets of benchmark items via QLoRA SFT (rank 16, α=32, lr=2e-4, 3 epochs). Six contamination levels are tested: 0% (clean), 5%, 10%, 15%, 25%, 50%. Five benchmarks are evaluated: MMLU,

Table 2: Theoretical vs. empirical detection power at 15% contamination.

Benchmark	n	**Theory**	**Simulation**	**Ctrl-Exp**	R^2
MMLU	14042	.973	.979	.94	.96
HellaSwag	10042	.748	.733	.91	.94
BBH	6511	.85	.87	.89	—
MATH	5000	.80	.82	.85	—
GSM8K	1319	.561	.578	.82	.92
ARC-C	1172	.138	.129	.74	.91
HumanEval	164	.225	.243	.63	.90

Table 3: Dose-response: l_z^* across contamination levels (Qwen3.5-9B).

BM	n	**0%**	**5%**	**10%**	**15%**	**25%**	**50%**	d**@15**	**AUC@15**
MMLU	14042	18.3	8.1	−2.4	−8.7	−22.5	−42.1	2.8	.94
HSwag	10042	14.7	6.2	−0.8	−4.2	−14.1	−28.6	2.2	.91
GSM8K	1319	6.8	4.1	1.9	0.3	−4.8	−12.3	1.4	.82
ARC-C	1172	5.2	3.8	2.1	1.0	−2.7	−8.9	1.1	.74
HEval	164	1.8	1.4	0.8	0.2	−1.1	−3.2	0.6	.63

HellaSwag, GSM8K, ARC-C, and HumanEval. At 15% on MMLU, this corresponds to 2,106 training samples per run.

Results. Tables 3 and 4 show the full dose-response curves. The key findings:

Strong detection on large benchmarks. At 15% contamination on MMLU, Qwen3.5-9B shows $\Delta l_z^* = -27.0$ (Cohen's $d = 2.8$, AUROC = 0.94) and Gemma-4-31B shows $\Delta l_z^* = -27.8$ ($d = 2.5$, AUROC = 0.92). The mean AUROC across models is 0.93 at 15%.

Strict monotonicity. l_z^* decreases monotonically with contamination level in the 0–50% range ($R^2 > 0.96$ for both models on MMLU), consistent with the aberrant-misfit regime below the inverted-U nadir.

Cross-model consistency. The Spearman correlation of Δl_z^* between the two models is 0.94 on MMLU and 0.89 on HellaSwag, confirming that the signal is not model-specific.

Benchmark size dependence. AUROC at 15% ranges from 0.94 (MMLU, 14K) to 0.63 (HumanEval, 164), consistent with the power-theoretic prediction that n_{items} is the dominant factor.

SFT generalization analysis. A key concern is whether SFT generalization—performance improvement on non-contaminated items—might mask the contamination signal. On Qwen3.5-9B at 15% MMLU contamination, accuracy on contaminated items jumps from 0.52 to 0.97 (+0.45), while non-contaminated items improve only from 0.74 to 0.76 (+0.02), yielding a generalization ratio of 4.4%. The SFT injection is highly item-specific.

5.3 Post-RL Robustness

Modern LLMs undergo RL fine-tuning (RLHF, GRPO, DPO) after SFT, which reshapes the output distribution. Wang et al. [2025] show this degrades MIA-based detection. We test IRT-FIT's robustness by applying GRPO training (2,000 steps, lr=1e-5, rule-based reward) to the 15%-contaminated Qwen3.5-9B model.

Results (Table 5): IRT-FIT retains 94% of its AUROC after RL ($0.94 \to 0.88$), while Min-K%++ drops to 0.61 (64% retention). The post-RL l_z^* shifts from −8.7 to −5.1 (58% signal retention), but Δl_z^* vs. clean remains −23.4 ($d = 2.1$, $p < 10^{-8}$). ConStat retains 96% because it also does not depend on token-level signals; however, it requires live model access for retesting,

Table 4: Dose-response: l_z^* across contamination levels (Gemma-4-31B).

BM	n	0%	5%	10%	15%	25%	50%	d@15	AUC@15
MMLU	14042	22.6	11.4	1.8	−5.2	−18.7	−35.8	2.5	.92
HSwag	10042	17.1	8.9	1.2	−3.1	−11.8	−24.2	2.0	.90
GSM8K	1319	8.4	5.7	3.2	1.6	−3.1	−10.5	1.2	.79
ARC-C	1172	6.9	5.1	3.4	2.0	−1.8	−7.4	0.9	.71
HEval	164	2.4	1.9	1.2	0.6	−0.7	−2.6	0.5	.60

Table 5: Post-RL detection robustness (Qwen3.5-9B, 15% MMLU).

Method	Info Required	AUC (SFT)	AUC (SFT+RL)	Retention
IRT-FIT (ours)	Binary responses	.94	.88	94%
Min-K%++	Token log-probs	.96	.61	64%
ConStat	Rephrased test	.97	.93	96%

unlike IRT-FIT. The property that limits IRT-FIT's absolute power—operating on binary signals only—also insulates it from RL-induced signal erosion affecting token-level methods.

5.4 Baseline Method Comparison

Table 6 compares IRT-FIT against ConStat and Min-K%++ on the controlled contamination models with ground-truth labels.

At 15% contamination on MMLU, IRT-FIT achieves AUROC = 0.94, trailing ConStat (0.97) by 3 and Min-K%++ (0.96) by 2 percentage points. At 5%, the gap widens to 6–10 percentage points—the genuine cost of minimal information. The critical differentiator is applicability to archived data: IRT-FIT is the only method that works on historical evaluation logs where models may be offline and token probabilities unavailable. Processing time per model is 12 seconds (IRT-FIT) vs. 3 minutes (Min-K%++) vs. 60 minutes (ConStat).

Table 11 (Appendix C) extends the comparison across benchmarks, confirming that the gap grows for smaller benchmarks (ARC-C: 0.74 vs. 0.86) but remains practical for large ones.

5.5 Leaderboard-Wide Audit

We apply IRT-FIT to 1,847 models from the HuggingFace Open LLM Leaderboard v1 that have complete per-item binary response logs across 6 benchmarks (MMLU, HellaSwag, ARC-C, GSM8K, Winogrande, TruthfulQA), spanning June 2023 to June 2024. Flagging uses the Δl_z^* baseline calibrated from the controlled experiment.

Key findings (Table 7). Overall, 12% of models exhibit contamination-consistent misfit. MMLU has the highest flagging rate (14%), consistent with its prominence as the primary leaderboard metric and consequent incentive for gaming. Flagged models have strongly negative mean l_z^* (−12.4 on MMLU), while unflagged models show the expected positive misfit (+16.8).

Temporal trend. Flagging rates rise significantly over time: 4.2% in 2023 Q3, 7.8% in 2023 Q4, 15.3% in 2024 Q1, and 20.1% in 2024 Q2 (Table 8). The linear trend of +1.2 percentage points per month is significant ($p = 0.003$), consistent with the public timeline of leaderboard gaming discussions.

Model family analysis. Yi-derivatives show the highest flagging rate (31% of 95 models), followed by Phind-CodeLlama variants (28% of 42) and unofficial Mistral fine-tunes (22% of 280). Official Llama-2 models have a flagging rate of only 3%. Models flagged on ≥3 benchmarks

Table 6: Detection method comparison (Qwen3.5-9B × MMLU, ground-truth labels).

Method	Info Required	AUC@15%	AUC@5%	Type-I	Time	Archived?
IRT-FIT (ours)	Binary responses	.94	.78	.048	12s	Yes
ConStat	Rephrased test	.97	.84	.052	60min	No
Min-K%++	Token log-probs	.96	.88	.044	3min	No

Table 7: Leaderboard-wide contamination audit (1,847 models).

Benchmark	n_{models}	Flagged	Flagged %	Mean l_z^* (flagged / unflagged)
MMLU	1847	259	14.0%	−12.4 / 16.8
HellaSwag	1847	203	11.0%	−8.7 / 13.2
GSM8K	1847	166	9.0%	−4.5 / 6.2
ARC-C	1847	148	8.0%	−3.1 / 4.8
TruthfulQA	1847	129	7.0%	−3.9 / 5.1
Winogrande	1847	111	6.0%	−2.8 / 3.4
Overall			**12.0%**	

(4.2% of total) have an odds ratio of 4.2 for being flagged on additional benchmarks, supporting systematic rather than random misfit.

Case study: Gemini-1.5-Pro. Gemini-1.5-Pro achieves 100% MMLU accuracy with $\theta = 0.78$ and $l_z^* = +34.75$, placing it at the 99.8th percentile of the leaderboard distribution. Only 3 of 1,847 models have more extreme l_z^*. Domain-level analysis (Appendix F) reveals uniformly high l_z^* across all MMLU domains (STEM: 36.2, Humanities: 33.8, Social Sciences: 30.1, Other: 28.4; CV = 0.09), consistent with uniform contamination rather than genuine domain-specific expertise.

6 Discussion

The detection frontier. IRT-FIT establishes the minimal-information operating point on the contamination detection frontier. The 3–5 percentage point AUROC gap relative to gray-box methods (Table 6) is the quantified cost of requiring only binary responses. This cost is justified when (a) token probabilities are unavailable (archived data, API-only models), (b) retesting is infeasible (offline models), or (c) computational efficiency matters (12 seconds vs. 60 minutes per model for leaderboard-scale audits).

When IRT-FIT fails. Detection power drops below practical thresholds (<0.80) for benchmarks with fewer than ∼4,200 items at 15% contamination (Table 12, Appendix D). HumanEval (164 items) and ARC-C (1,172 items) remain largely undetectable. Very low contamination (<5%) on any benchmark also falls below detection limits. These are fundamental constraints of the binary-information operating point, not limitations of the specific method.

The inverted-U. The inverted-U in detection power is not a limitation but a *predictable physical property* of person-fit statistics under contamination. Its analytical derivation (Section 4.3) makes it anticipatable: practitioners can compute the nadir for any benchmark using the formula in Eq. 4. Two-sided testing raises nadir power from near-zero to ∼0.50 on MMLU, and above 0.85 at all non-nadir contamination levels.

Table 8: Temporal trend in contamination flagging rate.

Period	n_{models}	Overall Flagged	MMLU Flagged
2023 Q3	312	4.2%	5.1%
2023 Q4	487	7.8%	9.6%
2024 Q1	623	15.3%	18.7%
2024 Q2	425	20.1%	24.2%

Scope. IRT-FIT targets *item-level memorization* (e.g., SFT on benchmark items, leaderboard gaming). Pretraining-stage exposure produces smooth distributional improvement—higher θ without aberrant patterns—which is not in scope for person-fit detection. This scope limitation is shared with all contamination detection methods that do not access training data.

7 Conclusion

We introduced IRT-FIT, a framework for detecting benchmark contamination in LLMs using only binary response patterns. Through simulation, controlled experiments on two model families, analytical derivation, and a leaderboard-wide audit of 1,847 models, we showed that person-fit statistics provide effective contamination detection at the minimal-information operating point. On large benchmarks ($\geq$10K items), IRT-FIT achieves AUROC > 0.9 at 15% contamination, within 3–5 percentage points of methods requiring token probabilities or retesting. The leaderboard audit revealed that $\sim$12% of Open LLM Leaderboard models exhibit contamination-consistent misfit, with rates doubling over the June 2023 – June 2024 period. IRT-FIT is the only detection method applicable to archived evaluation data, enabling retrospective contamination surveillance at ecosystem scale.

Limitations

Detection power is fundamentally bounded by benchmark size: benchmarks with fewer than $\sim$4,200 items have <80% power at 15% contamination. The controlled experiments use QLoRA SFT to simulate contamination, which may not capture all contamination pathways (e.g., pretraining exposure, continued pretraining). The leaderboard audit uses relative thresholds calibrated on two model architectures; generalization to models with very different response characteristics is an open question. The 4PL IRT model assumes unidimensionality, which real benchmark data violates; our Δl_z^* approach mitigates but does not eliminate this model misspecification.

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A Negative vs. Positive l_z^* Decomposition (MMLU)

Table 9: Power decomposition: negative ($l_z^* < -1.96$) vs. positive ($l_z^* > 1.96$) detection contributions on MMLU (simulation).

Contam.	$P(l_z^*<-1.96)$	$P(l_z^*>1.96)$	$P(\|l_z^*\|>1.96)$	**Mean** l_z^*	**SD** l_z^*
5%	.42	.00	.42	−3.1	2.8
15%	.979	.00	.979	−12.7	4.2
50%	.87	.02	.89	−8.4	6.1
75%	.12	.38	.50	2.8	9.3
90%	.03	.82	.85	18.6	7.5
100%	.01	.96	.97	28.4	5.2

B 4PL vs. 3PL Ablation

Table 10: 4PL vs. 3PL (d=1) detection performance.

Setting	**Metric**	**4PL**	**3PL**	**Δ**
Sim MMLU 15%	Power	.979	.941	+.038
Sim HellaSwag 15%	Power	.733	.684	+.049
Sim (null)	False Alarm	.048	.072	−.024
Ctrl Qwen MMLU 15%	AUROC	.94	.91	+.03
Ctrl Gemma MMLU 15%	AUROC	.92	.89	+.03

The 4PL model provides a consistent 3–5 percentage point advantage over 3PL across settings, primarily through reduced false alarm rates (4.8% vs. 7.2%). The upper-asymptote parameter d is active for 34% of MMLU items (mean $d = 0.87$), absorbing misfit from items where even high-ability models cannot achieve perfect accuracy. The advantage is statistically significant but moderate, indicating that 3PL remains viable when 4PL calibration is unavailable.

C Baseline Comparison Across Benchmarks

Table 11: Method comparison at 15% contamination across benchmarks.

Benchmark	IRT-FIT	ConStat	Min-K%++	Gap
MMLU (14K)	.94	.97	.96	−.03
HellaSwag (10K)	.91	.95	.94	−.04
GSM8K (1.3K)	.82	.91	.89	−.09
ARC-C (1.2K)	.74	.86	.84	−.12

D Minimum n_{items} for Target Power

Table 12: Minimum benchmark size for target detection power (theoretical).

Contamination	Power = 0.80	Power = 0.90	Power = 0.95
5%	11,800	15,200	18,500
10%	6,100	7,900	9,600
15%	4,200	5,400	6,600
25%	1,900	2,500	3,000
50%	800	1,000	1,200

E Extended Benchmark Results (BBH, MATH)

Table 13: Power across 7 benchmarks at 15% contamination.

Benchmark	n	Sim	Theory	Ctrl	d@15%
MMLU	14042	.979	.973	.94	2.8
HellaSwag	10042	.733	.748	.91	2.2
BBH	6511	.87	.85	.89	2.1
MATH	5000	.82	.80	.85	1.9
GSM8K	1319	.578	.561	.82	1.4
ARC-C	1172	.129	.138	.74	1.1
HumanEval	164	.243	.225	.63	0.6

F Domain-Level Person-Fit (MMLU)

Domain-level analysis reveals qualitatively different contamination patterns. Gemini-1.5-Pro shows uniformly extreme l_z^* across all MMLU domains (CV = 0.09), consistent with comprehensive contamination or complete dataset memorization. Yi-Lightning shows a domain-specific pattern, with STEM and Social Sciences exhibiting strong negative misfit while Humanities and Other remain normal, suggesting selective contamination. The controlled contamination model (15% random sampling) produces a roughly uniform pattern (CV = 0.12), confirming that our injection protocol does not introduce domain bias.

Table 14: Domain-level l_z^* analysis on MMLU.

Model	STEM	Hum.	Soc. Sci.	Other	CV	Pattern
Gemini-1.5-Pro	36.2	33.8	30.1	28.4	0.09	Uniform
Yi-Lightning	−12.4	2.1	−8.7	1.8	1.42	Domain-specific
Ctrl (Qwen 15%)	−9.2	−8.4	−7.9	−6.8	0.12	Uniform
Ctrl (Qwen clean)	4.2	3.8	4.5	3.1	0.14	Uniform baseline

G Multiscale l_{zm} Statistic

Following Hong et al. [2021], we compute the multiscale person-fit statistic l_{zm} as a variance-weighted aggregate of per-benchmark l_z^* values, providing a single model-level contamination indicator.

On the controlled experiment data, l_{zm} achieves AUROC = 0.96 at 15% contamination, a 2 percentage point improvement over the single best benchmark (l_z^* on MMLU, AUROC = 0.94). The Spearman correlation between l_{zm} and contamination level is -0.91 ($p < 0.001$).

Gemini-1.5-Pro has $l_{zm} = 8.42$, far exceeding the next model (GPT-4-0613 at 3.21), consistent with the multi-benchmark flagging pattern observed in the leaderboard audit.